

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BTECH/IMSC

SEMESTER: I/ADD

BRANCH: BT/CIVIL/CHEMICAL/MECH/PIE/FT/PHYSICS

SESSION : MO/2024

SUBJECT: CS24101 PROGRAMMING FOR PROBLEM SOLVING

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

		CO	BL
Q.1(a)	What is the difference between a syntax error and logical error? Give a suitable example for explanation.	[2] CO3	BL2
Q.1(b)	Write an algorithm and draw a flowchart to find sum of series 1+3+5+.... +N, Where N is positive odd Integer.	[3] CO1	BL4
Q.2(a)	Briefly explain the four basic or primary data types in C?	[2] CO2	BL2
Q.2(b)	What is the limitation of conditional operator? Using conditional operator write a C program to find largest of three numbers?	[3] CO3	BL3
Q.3(a)	What is the use of <i>comma</i> and <i>sizeof</i> () operator? Give suitable example for explanation?	[2] CO2	BL3
Q.3(b)	Write a C program to find roots of quadratic equation ($ax^2 + bx + c = 0$).	[3] CO4	BL5
Q.4(a)	Give suitable explanation when you prefer <i>while loop</i> compared to <i>for loop</i> ?	[2] CO2	BL3
Q.4(b)	Write a C-program to find sum of the digits of positive integer and check if the number is palindrome or not?	[3] CO3	BL4
Q.5(a)	Write a C program to find the sum and sum of squares of the first 15 positive natural numbers?	[2] CO3	BL3
Q.5(b)	Write down the value of x for each j in the code given below:	[3] CO3	BL5
<pre>main () { int j, x =0; for (j = 0; j<=5; j++) { switch (j - 1) { case 0: case -1 : x = x + 1; break; case 1: case 2: case 3: x = x + 2; break; default: x = x + 3; } printf("x=%d\n", x); } }</pre>			